Nikhil Kumar Thota

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Education

Ph.D. Chemical and Biomolecular Engineering Jan 2026
Johns Hopkins University, Baltimore, MD

M.S.E Chemical and Biomolecular Engineering Dec 2021
Johns Hopkins University, Baltimore, MD

B.Tech. Chemical Engineering May 2019
Manipal Institute of Technology, Manipal, India

Research Experience

Postdoctoral Fellow Sep 2026 – Present

- Working in Dr. Pavel Jungwirth's group at IOCB, Prague on scaling Bayesian Force Fields to larger biomolecules.
- Currently focusing on intrinsically disordered proteins and ion transporters.

Ph.D. Candidate Jan 2022 – Jan 2026

Advisors: Dr. Rigoberto Hernandez (Johns Hopkins), Dr. Paulette Clancy (Johns Hopkins)

- Developed nested autoencoders (NestedAE) and nested heteroencoders (NestedHE) to construct low dimensional material embeddings for materials discovery from high dimensional and multi-scale datasets.
- Developed a multi-agent reinforcement learning (MARL) algorithm to construct sparse feature subsets.

Ph.D. Intern Jul 2024 – Sep 2024

Advisor: Dr. Jinhui Tao (PNNL, Richland), Dr. Rigoberto Hernandez (Johns Hopkins)

- Trained bayesian neural networks to predict binding free energies of polypeptoid sequences to calcite surfaces.
- Mentored Bryan Zhan, a highschooler, on fundamentals of calculus and introductory data analysis using Python.

M.S.E Student Sep 2020 – Dec 2021

Advisor: Dr. Rigoberto Hernandez (Johns Hopkins)

- Optimized cost of running Steered Molecular Dynamics (SMD) simulations for unfolding proteins by co-developing a variable-sized solvent box method.
- Automated running SMD simulations on Multi Core CPU and GPU clusters using Bash and Python scripts.

Undergraduate Intern Jan 2019 – Dec 2019

Advisor : Dr. Bhushan Toley (Indian Institute of Science, Bengaluru, India)

- Identified reaction products and pathways involved in loop mediated and isothermal amplification (LAMP) of DNA used in PCR free diagnosis kits of tuberculosis and COVID-19.
- Developed an algorithm to simulate the devised stoichiometric model and validated model outputs using experimental gel electrophoresis images and fluorescence readouts.

Publications

1. **Thota, N. K.**; Hernandez, R. Nested Autoencoders and Nested Heteroencoders for Multi-scale Property Prediction of Perovskite Solar Cells and Photocatalysts (in preparation)
2. Baird, S.; ... **Thota, N. K.**; ... Zou, Y. Bayesian Optimization Hackathon for Chemistry and Materials. *ChemRxiv*, June 23, 2025. DOI: 10.26434/chemrxiv-2023-tkcd9
3. Priyadarshini, M. S.*; **Thota, N. K.***; Hernandez, R. ReLMM: Reinforcement Learning Optimizes Feature Selection in Modeling Materials. *J. Chem. Inf. Model.* **2025**, 65 (1), 153–161. DOI: 10.1021/acs.jcim.4c01934
4. **Thota, N. K.***; Priyadarshini, M. S.*; Hernandez, R. NestedAE: Interpretable Nested Autoencoders for Multi-Scale Materials Characterization. *Mater. Horiz.* **2024**, 11 (3), 700–707. DOI: 10.1039/D3MH01484C.
5. **Thota, N. K.**; Quirk, S.; Zhuang, Y.; Stover, E. R.; Lieberman, R. L.; Hernandez, R. Correlation between Chemical Denaturation and the Unfolding Energetics of Acanthamoeba Actophorin. *Biophysical Journal* **2022**, S0006349522038590. DOI: 10.1016/j.bpj.2022.11.2941
6. Zhuang, Y.; **Thota, N. K.**; Quirk, S.; Hernandez, R. Implementation of Telescoping Boxes in Adaptive Steered Molecular Dynamics. *J. Chem. Theory Comput.* **2022**, 18 (8), 4649–4659. DOI: 10.1021/acs.jctc.2c00498

7. Kaur, N.*; **Thota, N. K.***; Toley, B. J. A Stoichiometric and Pseudo Kinetic Model of Loop Mediated Isothermal Amplification. *Comput. Struct. Biotechnol. J.* **2020**, 18, 2336–2346. DOI: 10.1016/j.csbj.2020.08.020

* indicates equal contribution

Conference Presentations

1. **Thota, N. K. (Presenter)**, Priyadarshini, M. S., Hernandez, R *Multiscale Property Prediction and Design of Perovskites Using Nested Autoencoders*. Presented at the American Institute of Chemical Engineers, Boston, MA, Nov 5, 2025 (Oral)
2. **Thota, N. K. (Presenter)**, Hernandez, R *Multi-Task Learning Using Nested Autoencoders for Designing Perovskites*. Presented at the Johns Hopkins Data Science and AI Institute Symposium, Baltimore, MD, Sep 16, 2025 (Poster)
3. **Thota, N. K. (Presenter)**, Priyadarshini, M. S., Hernandez, R *Multiscale Modeling of Materials Using Nested Autoencoders*. Presented at the AI-X Foundry Symposium, Johns Hopkins University, Baltimore, MD, Sep 27, 2023 (Poster)
4. **Thota, N. K. (Presenter)**, Quirk, S.; Zhuang, Y.; Hernandez, R. *Mutational Assay of an Actophorin Protein Using Adaptive Steered Molecular Dynamics*, Presented at American Chemical Society Conference, Chicago, IL, Aug 22, 2022 (Oral)

Teaching Experience

Instructor for Software Carpentry Sep 2024 – Dec 2024

Johns Hopkins University, Baltimore, MD

- Designed and taught a course on introductory python programming and applications to 30 graduate students.
- Managed a team of a graduate TA and grader to handle office hours and grading.

Teaching Assistant for Process Dynamics and Control Sep 2022 – Dec 2022

Johns Hopkins University, Baltimore, MD

- Graded undergraduate weekly assignments and tutored 17 undergraduates during weekly recitation sessions.
- Gave lectures on applications of machine learning in process control.

Awards and Certificates

1. Awarded the Institute of Organic Chemistry and Biochemistry (IOCB) Postdoctoral Fellowship to conduct my postdoctoral research. Apr 2026

2. Department graduate TA award for Software Carpentry Apr 2025

3. Among 12 finalists selected for 2022 Empower Your Pitch (EYP) competition (3 minute research communication competition) Apr 2022

Academic Service and Leadership

Executive Board Member in Graduate Student Liason Committee (GSLC) Sep 2022 – Sep 2023

Johns Hopkins University, Baltimore, MD, USA

- Facilitated two way communication between graduate students in the department and the university wide Graduate Research Organization (GRO).
- Organized student led seminars for graduate students in the department.

Voluntary Services

- Reviewer for AICHE Journal, JCTC and ML4LMS Workshop (Part of ICML 2024).

Skills

ML Models : Autoencoders, Gaussian Processes, Diffusion models **ML Interpretability :** SHAP
Optimization Methods : Reinforcement Learning, Bayesian Optimization
Coding Languages (Packages) : Python (PyTorch, TensorFlow, NumPy, Scipy, SHAP, Pytest(Code testing), Setuptools (Python Packaging)), Bash, MATLAB (SimBiology, Parallel Computing), C++, HTML, Markdown **Version Control:** Git, GitHub
Simulation Software : NAMD, **Visualization Software :** VMD, PyMOL, OVITO
Enhanced Sampling : Steered Molecular Dynamics